

WARNEMUENDE, Germany

WMO: 101700

Lat: 54.1803N

Lon: 12.0808E

Elev: 13

StdP: 101.17

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-9.7	-7.2	-13.1	1.2	-9.3	-10.5	1.5	-6.7	15.8	3.6	14.6	2.9	3.3	110	0.642

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.3	26.5	18.2	24.3	17.7	22.2	17.1	19.5	24.0	18.6	22.7	17.8	21.4	3.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.9	12.9	21.2	17.0	12.1	20.5	16.2	11.5	19.7	55.7	24.1	52.7	22.9	50.1	21.5	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	12.4	10.9	DB -9.7	31.3	4.4	2.4	-12.9	33.0	-15.4	34.4	-17.9	35.7	-21.0	37.5		
(5)			WB -10.3	22.1	4.1	1.3	-13.2	23.0	-15.6	23.8	-17.9	24.5	-20.9	25.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	1.2	1.0	4.0	7.1	11.9	15.2	17.6	17.3	14.1	10.2	5.2	2.7
	DBStd	6.80	4.78	4.15	3.46	3.25	3.34	2.99	2.56	2.59	2.30	3.14	3.04	3.32
	HDD10.0	1238	274	254	187	100	16	1	0	0	1	36	145	226
	HDD18.3	3464	532	487	444	338	201	103	45	53	130	253	394	484
	CDD10.0	871	0	1	2	12	75	156	236	226	123	41	1	1
	CDD18.3	56	0	0	0	0	2	9	22	20	2	0	0	0
	CDH23.3	299	0	0	0	1	21	55	107	105	11	0	0	0
	CDH26.7	66	0	0	0	0	1	13	24	26	1	0	0	0
(14) Wind	WSAvg	4.7	5.4	4.9	4.7	4.4	4.0	4.4	4.8	4.3	4.9	4.7	5.0	5.3
(15) Precipitation	PrecAvg	601	47	33	39	39	49	64	71	64	54	45	49	50
	PrecMax	955	93	84	98	134	114	155	344	193	159	106	130	105
	PrecMin	419	0	3	6	1	4	0	4	9	12	7	5	10
	PrecStd	116	23	17	20	25	25	34	50	41	31	22	26	23
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.8	13.5	15.7	20.9	26.0	28.9	29.7	29.5	25.0	20.5	12.8	11.6
		MCWB	9.2	10.5	10.8	13.0	17.6	18.8	19.4	19.3	18.4	16.6	10.9	9.8
	2%	DB	8.9	10.2	11.8	16.7	22.6	24.5	25.9	26.2	21.5	17.8	11.5	10.2
		MCWB	7.7	7.9	8.5	11.3	15.3	17.7	18.4	18.3	15.9	14.3	10.1	8.8
	5%	DB	8.0	7.6	9.8	13.6	19.4	21.7	23.8	23.3	19.1	15.8	10.2	8.8
		MCWB	6.8	5.9	7.6	9.9	13.8	16.7	18.0	17.9	15.2	13.1	9.1	7.8
	10%	DB	6.7	6.0	8.3	11.7	16.9	19.8	21.9	21.2	17.6	14.2	9.2	7.2
		MCWB	5.5	4.6	6.5	8.8	12.8	15.8	17.5	17.0	14.6	12.1	8.1	6.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.5	10.8	11.3	14.0	17.7	19.8	21.1	20.9	19.0	17.3	11.4	10.2
		MCDB	10.5	13.0	15.0	19.6	24.1	25.3	25.8	26.6	23.9	20.1	12.3	11.1
	2%	WB	7.8	8.2	9.2	11.7	16.0	18.4	19.8	19.2	17.2	15.0	10.2	8.9
		MCDB	8.5	10.0	11.2	15.7	20.6	22.9	24.0	23.5	20.1	17.0	11.1	10.0
	5%	WB	6.9	6.2	7.9	10.2	14.4	17.4	18.7	18.4	16.0	13.5	9.2	7.7
		MCDB	7.9	7.6	9.5	13.3	18.7	21.1	22.7	22.0	18.1	15.3	10.2	8.8
	10%	WB	5.6	4.8	6.6	9.0	13.0	16.3	17.9	17.5	14.9	12.4	8.3	6.3
		MCDB	6.6	5.9	8.2	11.5	16.0	19.4	21.3	20.7	17.0	13.9	9.2	7.1
(35) Mean Daily Temperature Range	5% DB	MDBR	3.7	3.9	5.0	5.7	6.6	6.2	6.3	6.5	5.7	5.2	3.9	3.5
		MCDBR	4.5	5.6	7.4	9.7	11.5	10.2	10.4	10.8	8.8	7.5	4.8	4.9
	5% WB	MCWBR	4.2	4.6	5.3	5.8	5.8	5.3	5.2	5.2	5.1	4.9	4.4	4.6
		MCDWR	4.4	5.7	6.8	9.1	10.0	9.2	8.7	8.8	7.7	6.6	4.6	4.6
(40) Clear-Sky Solar Irradiance	taub		0.319	0.341	0.372	0.400	0.393	0.391	0.410	0.409	0.377	0.362	0.339	0.311
		taud	2.315	2.318	2.263	2.255	2.332	2.362	2.327	2.342	2.412	2.407	2.365	2.306
	Edn at Noon		627	741	796	825	853	858	831	809	789	711	598	553
		Edh at Noon	58	80	104	120	117	115	117	109	90	73	56	48
(44) All-Sky Solar Radiation	RadAvg		0.52	1.15	2.41	4.11	5.22	5.56	5.17	4.22	3.05	1.60	0.66	0.37
	RadStd		0.07	0.19	0.23	0.51	0.59	0.40	0.63	0.31	0.30	0.21	0.08	0.03

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (55 neighbors)	+0.38	N/A	N/A	N/A	N/A	N/A	N/A	-156	N/A	N/A

Nomenclature: See separate page